Penguins – UMAP Analysis

Equipo 7
Integrantes:
Fajardo Zavala Yendali Viridiana
González Rivas José Rafael
Maldonado Zamora Silvia Viridiana
Olivera Flores Jonathan
Valencia Quiroz Luis Roberto

2025-12-10

Load Data

```
df <- read.csv("C:/Users/usuario/OneDrive/Desktop/Proyecto 2 Diplomado/Penguins/Club_Penguin/penguins.c

dim(df)
```

```
## [1] 344   8
```

```
glimpse(df)
```

```
## Rows: 344
## Columns: 8
## $ species           <chr> "Adelie", "Adelie", "Adelie", "Adelie", "Adelie", "A~
## $ island            <chr> "Torgersen", "Torgersen", "Torgersen", "Torgersen", ~
## $ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
## $ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
## $ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
## $ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
## $ sex               <chr> "male", "female", "female", NA, "female", "male", "f~
## $ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

```
head(df)
```

```
##   species    island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
## 1  Adelie Torgersen           39.1          18.7               181        3750
## 2  Adelie Torgersen           39.5          17.4               186        3800
## 3  Adelie Torgersen           40.3          18.0               195        3250
## 4  Adelie Torgersen             NA            NA                NA          NA
## 5  Adelie Torgersen           36.7          19.3               193        3450
## 6  Adelie Torgersen           39.3          20.6               190        3650
##      sex year
```

```
## 1   male 2007
## 2 female 2007
## 3 female 2007
## 4   <NA> 2007
## 5 female 2007
## 6   male 2007
```

Preprocessing

```
df1 <- df[,1:8]
colSums(is.na(df1))
```

```
##           species            island    bill_length_mm     bill_depth_mm
##                 0                 0                 2                 2
## flipper_length_mm       body_mass_g               sex              year
##                 2                 2                11                 0
```

```
df1c <- df1 %>%
  mutate(across(where(is.numeric), ~ ifelse(is.na(.), mean(., na.rm = TRUE), .)))

colSums(is.na(df1c))
```

```
##           species            island    bill_length_mm     bill_depth_mm
##                 0                 0                 0                 0
## flipper_length_mm       body_mass_g               sex              year
##                 0                 0                11                 0
```

```
# Factor conversion
df1c <- df1c %>%
  mutate(species = as.factor(species),
         island = as.factor(island),
         sex = as.factor(sex))

dim(df1c)
```

```
## [1] 344   8
```

```
glimpse(df1c)
```

```
## Rows: 344
## Columns: 8
## $ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
## $ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
## $ bill_length_mm    <dbl> 39.10000, 39.50000, 40.30000, 43.92193, 36.70000, 39~
## $ bill_depth_mm     <dbl> 18.70000, 17.40000, 18.00000, 17.15117, 19.30000, 20~
## $ flipper_length_mm <dbl> 181.0000, 186.0000, 195.0000, 200.9152, 193.0000, 19~
## $ body_mass_g       <dbl> 3750.000, 3800.000, 3250.000, 4201.754, 3450.000, 36~
## $ sex               <fct> male, female, female, NA, female, male, female, male~
## $ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

```r
head(df1c)
```

```
##   species    island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
## 1  Adelie Torgersen       39.10000      18.70000          181.0000    3750.000
## 2  Adelie Torgersen       39.50000      17.40000          186.0000    3800.000
## 3  Adelie Torgersen       40.30000      18.00000          195.0000    3250.000
## 4  Adelie Torgersen       43.92193      17.15117          200.9152    4201.754
## 5  Adelie Torgersen       36.70000      19.30000          193.0000    3450.000
## 6  Adelie Torgersen       39.30000      20.60000          190.0000    3650.000
##      sex year
## 1   male 2007
## 2 female 2007
## 3 female 2007
## 4   <NA> 2007
## 5 female 2007
## 6   male 2007
```

```r
count(df1c, species)
```

```
##     species   n
## 1    Adelie 152
## 2 Chinstrap  68
## 3    Gentoo 124
```

```r
count(df1c, island)
```

```
##      island   n
## 1    Biscoe 168
## 2     Dream 124
## 3 Torgersen  52
```

```r
count(df1c, sex)
```

```
##      sex   n
## 1 female 165
## 2   male 168
## 3   <NA>  11
```

UMAP Básico

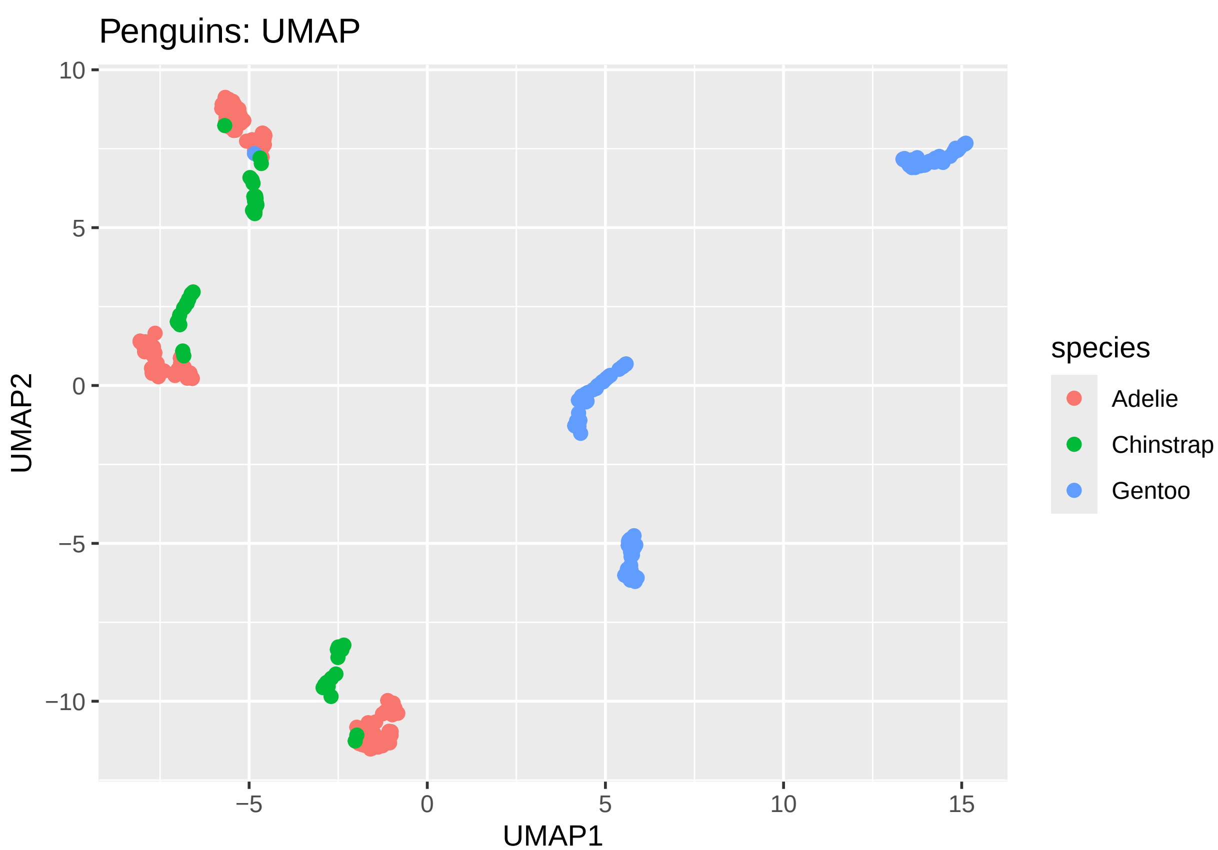
```r
df_umap <- df1c %>%
  select(where(is.numeric)) %>%
  scale() %>%
  umap()

str(df_umap)
```

```
##  num [1:344, 1:2] -1.59 -1.73 -1.9 -2.71 -1.01 ...
##  - attr(*, "scaled:center")= num [1:2] 1.084 -0.313
```

```r
datos <- data.frame(UMAP1=df_umap[,1], UMAP2=df_umap[,2], species=df$species)
head(datos)
```

```
##        UMAP1      UMAP2 species
## 1 -1.5855041 -10.841814  Adelie
## 2 -1.7336826 -11.015949  Adelie
## 3 -1.9020954 -11.292904  Adelie
## 4 -2.7136777  -9.302032  Adelie
## 5 -1.0148084 -11.078438  Adelie
## 6 -0.9169847 -10.377369  Adelie
```

```r
# Plot
ggplot(datos, aes(x = UMAP1, y = UMAP2, color = species)) +
  geom_point(size = 2) +
  labs(title="Penguins: UMAP")
```



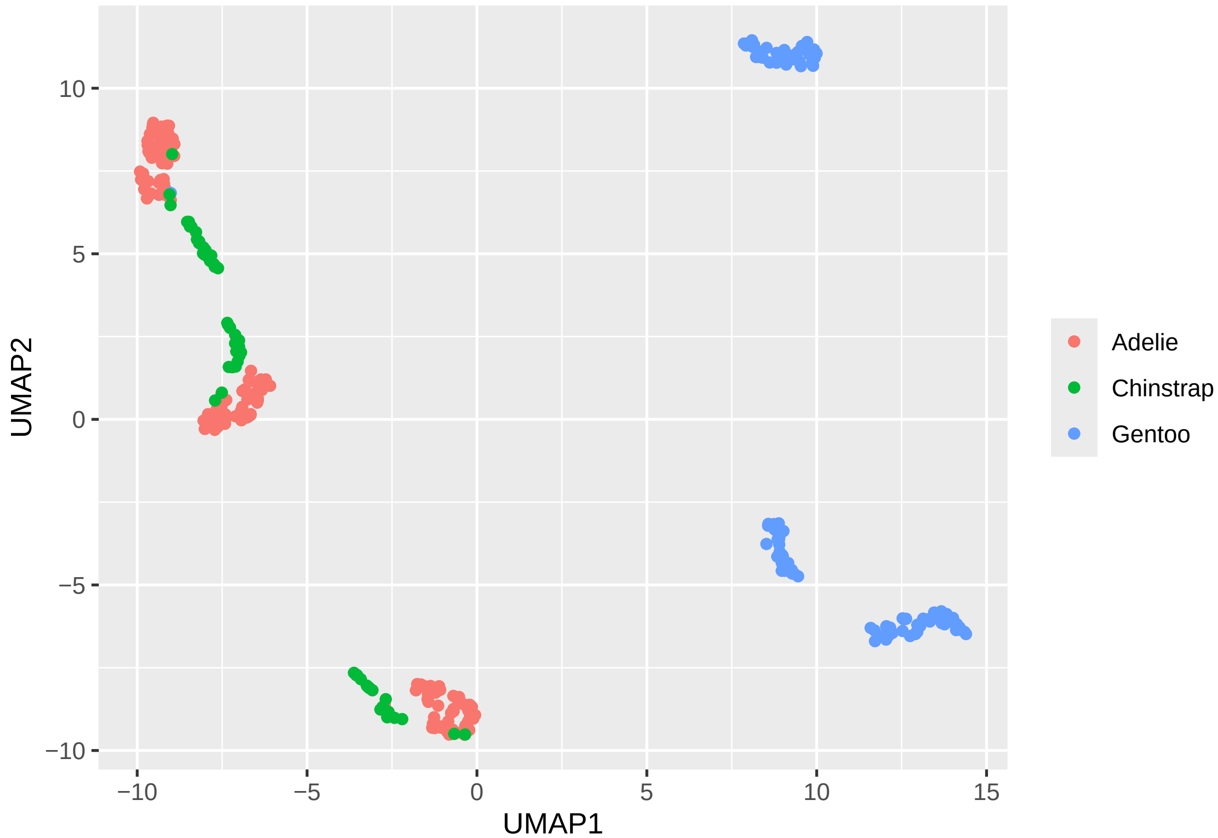
UMAP via tidymodels

```r
df1c <- df1c %>% select(-sex, -island)

umap_rec <- recipe(~., data=df1c) %>%
  update_role(species, new_role="id") %>%
```

```r
  step_normalize(all_predictors()) %>%
  step_umap(all_predictors())

umap_res <- prep(umap_rec)
umap_res
```

```
##

## -- Recipe ----------------------------------------------------------------------------------------------

##

## -- Inputs

## Number of variables by role

## predictor: 5
## id:        1

##

## -- Training information

## Training data contained 344 data points and no incomplete rows.

##

## -- Operations

## * Centering and scaling for: bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g, year | T

## * UMAP embedding for: bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g, year | Trained
```

```r
juice(umap_res) %>%
  ggplot(aes(UMAP1, UMAP2)) +
  geom_point(aes(color=species), size=1.5) +
  labs(color=NULL)
```

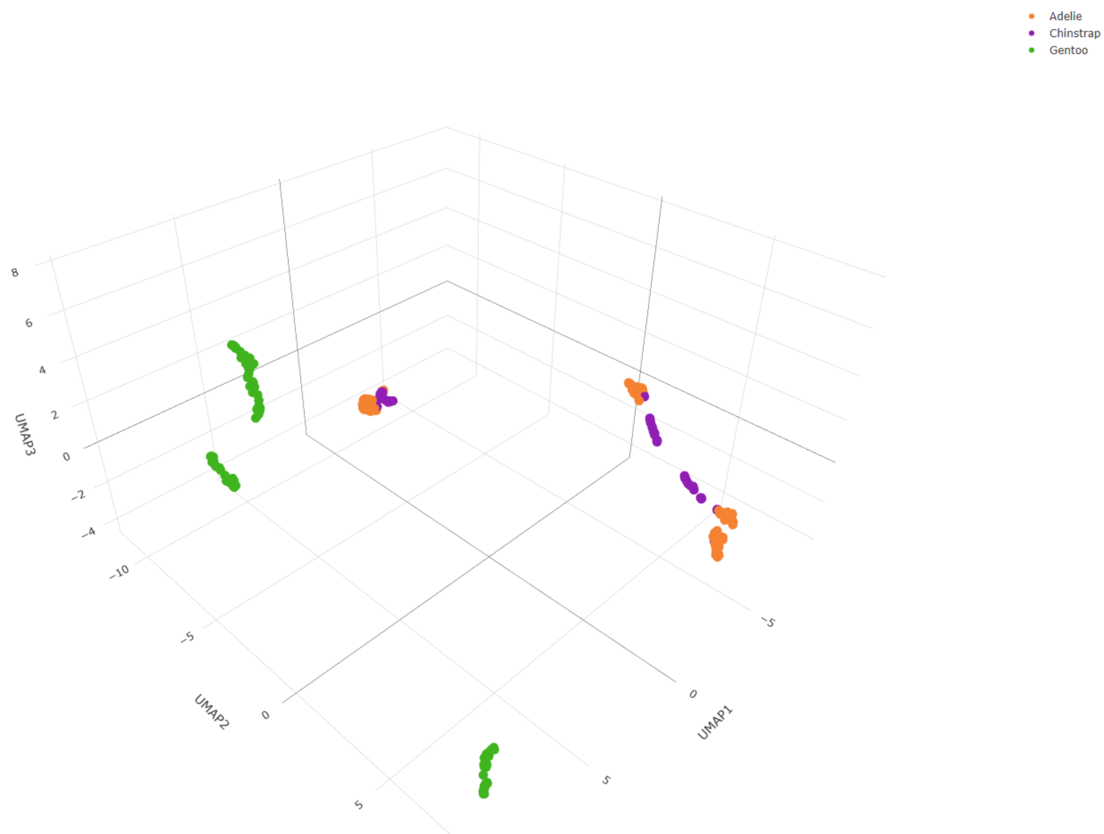


3D UMAP

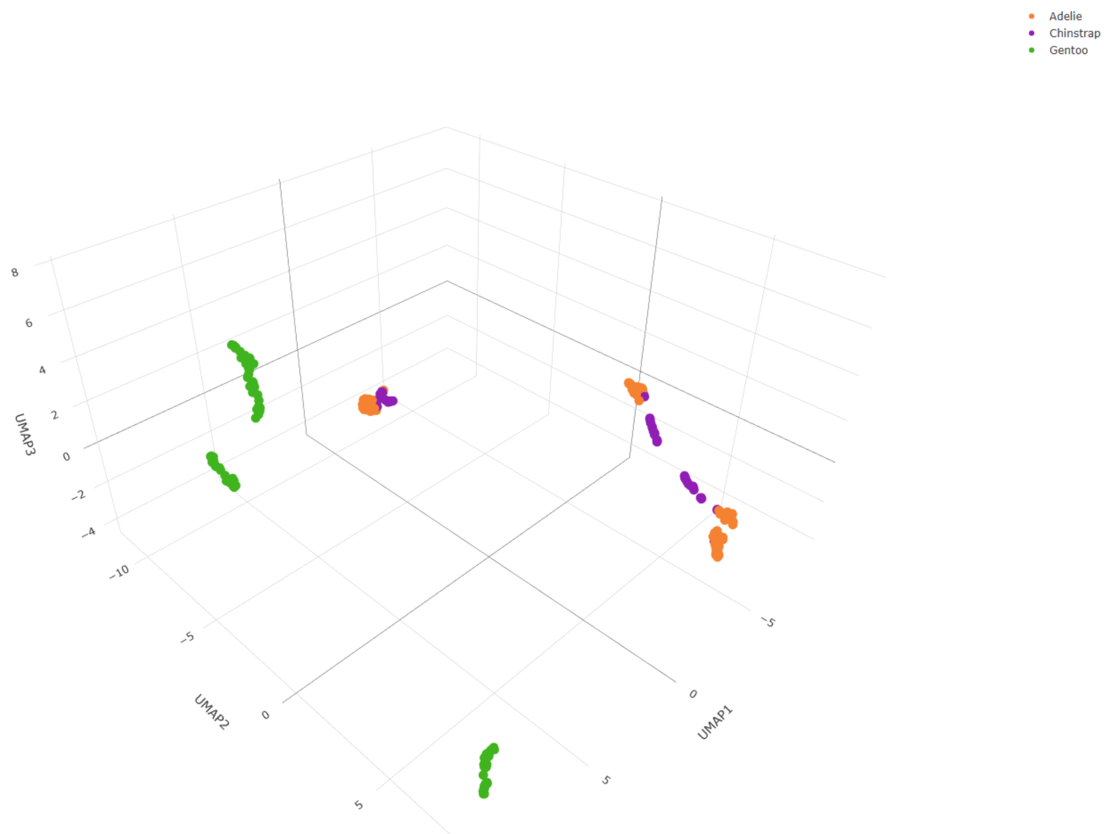
```
library(plotly)
library(webshot2)
library(htmlwidgets)
library(knitr)

# ---- Run your UMAP 3D code ----
df_umap3 <- df1c %>%
  select(where(is.numeric)) %>%
  scale() %>%
  umap(n_components = 3)

umap_df3 <- data.frame(
  UMAP1 = df_umap3[,1],
  UMAP2 = df_umap3[,2],
  UMAP3 = df_umap3[,3],
  species = factor(df$species)
)

colors <- c("#F58231", "#911EB4", "#3fb41e")

hover_text <- paste(
```

```r
  "species:", umap_df3$species,
  "Dimension 1:", round(umap_df3$UMAP1, 3),
  "Dimension 2:", round(umap_df3$UMAP2, 3),
  "Dimension 3:", round(umap_df3$UMAP3, 3)
)

p <- plot_ly(
  data = umap_df3,
  x = ~UMAP1, y = ~UMAP2, z = ~UMAP3,
  type = "scatter3d", mode = "markers",
  marker = list(size = 6),
  text = hover_text, hoverinfo = "text",
  color = ~species, colors = colors
)

# Save temporary HTML file
htmlwidgets::saveWidget(p, "umap3d_temp.html", selfcontained = TRUE)

# Take screenshot (requires Chrome or Chromium)
webshot2::webshot(
  "umap3d_temp.html",
  file = "umap3d_plot.png",
  vwidth = 1400, vheight = 1000
)
```

```
## file:///C:/Users/usuario/OneDrive/Desktop/Proyecto 2 Diplomado/Penguins/Club_Penguin/umap3d_temp.html
```



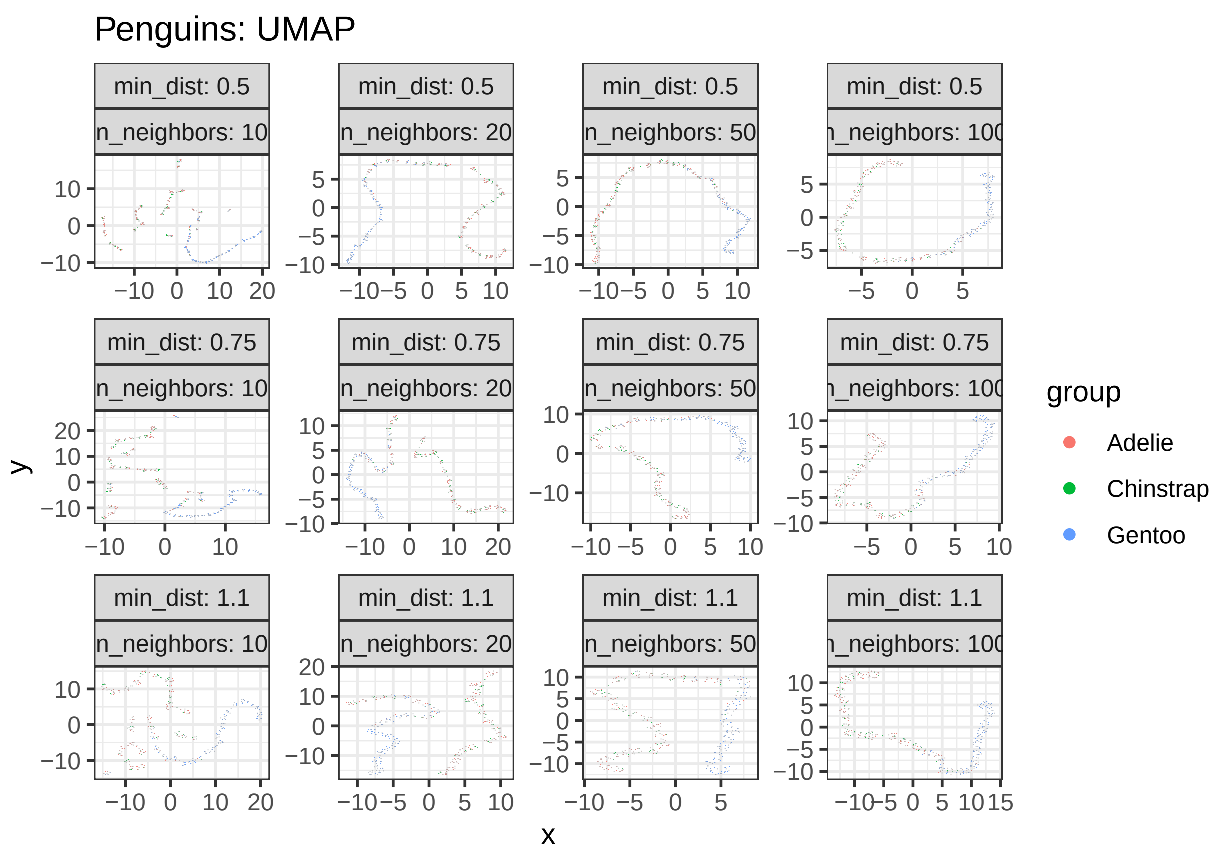

```r
# Insert screenshot into PDF
knitr::include_graphics("umap3d_plot.png")
```



Exploración de Hipereparámetros

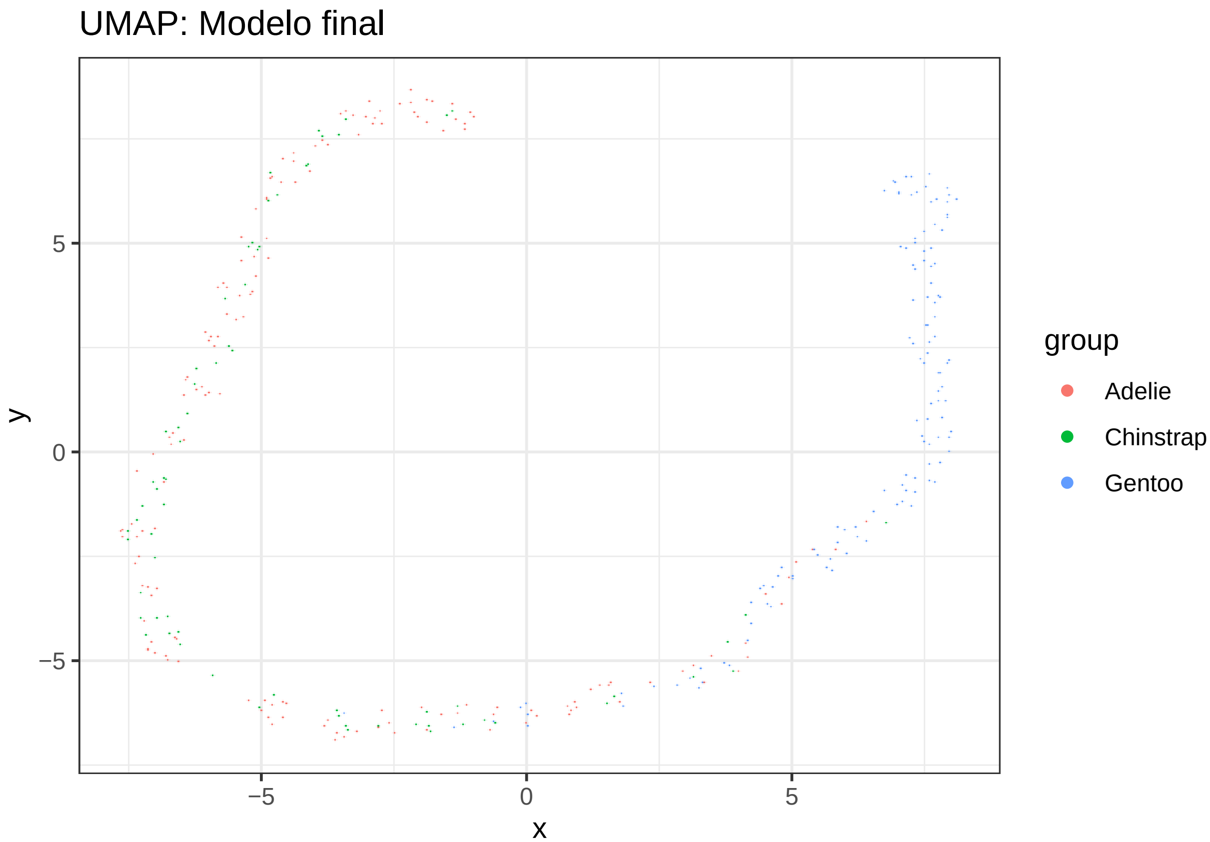
```
umap_params <- expand.grid(n_neighbors=c(10,20,50,100),
                           min_dist=c(0.5,0.75,1.1))

numeric_df <- df1c %>% select(where(is.numeric))

umaps <- lapply(seq(nrow(umap_params)), function(i) {
  umap(X=numeric_df,
       n_neighbors=umap_params$n_neighbors[i],
       min_dist=umap_params$min_dist[i])
})

d <- rbindlist(lapply(seq(nrow(umap_params)), function(i) {
  data.table(
    x=umaps[[i]][,1], y=umaps[[i]][,2],
    n_neighbors=umap_params$n_neighbors[i],
    min_dist=umap_params$min_dist[i],
    group=df$species
  )
}))
```

```
p <- ggplot(d) +
  geom_scattermore(aes(x=x, y=y, colour=group), pointsize=2) +
  theme(axis.text=element_blank(), axis.ticks=element_blank(),
        axis.title=element_blank(), legend.position="none") +
  facet_wrap(min_dist ~ n_neighbors, labeller=label_both, scales="free") +
  theme_bw() +
  labs(title="Penguins: UMAP")

p
```



Modelo Final

```
Final_plot <- d %>%
  filter(min_dist==0.5, n_neighbors==100) %>%
  ggplot(aes(x=x, y=y, col=group)) +
  geom_scattermore(pointsize=1.1) +
  theme(axis.text=element_blank(), axis.ticks=element_blank(),
        axis.title=element_blank(), legend.position="none") +
  theme_bw() +
  labs(title="UMAP: Modelo final")

Final_plot
```


UMAP: Modelo final
y
x
5
0
−5
−5
0
5
group
Adelie
Chinstrap
Gentoo